

get paid on their own—from credit or reserve tokens—and even energy usage alerts are generated. For instance, with open trading in electricity tokens, just like with other commodities, token rates also fluctuate. So, a smart energy agent can even instruct the microwave oven, a known high-energy guzzler, to avoid working during peak price hours!

Evolution of blockchain in the energy sector

At this year's *IEW* held at Bengaluru, Singh spoke about both opportunities and challenges when applying blockchain technology to the energy sector. He talked of how in the early days, around 2015, companies exploring the potential of blockchain tended to look for solutions within single organisations. But it soon emerged that the utility space needed groups or consortiums to work together as partners, on a common platform.

Using the oil sector as an example, Singh turned the spotlight on oil traders. From these traders' points of view, positioned in the middle of the extraction-to-market chain, there are innumerable transactions, involving asking oil producers for quotations, and placing orders directly with sellers and carriers like tanker fleet owners, etc. All these transactions need support from banks and regulatory authorities.

Further, these middlemen or traders then transact with buyers, who can be traders themselves, or governments and firms in the Big Oil league. These set of transactions require regulatory approvals, as oil is shipped across national and international boundaries, being subjected to tax and duties. These also require finance, which is usually in the form of credit.

With diverse firms operating on different systems, these transactions take an unnecessarily long time, leading to lost man hours, delayed deliveries and money lost during a slower business cycle.

Now, consortiums comprising every entity in the ecosystem—oil producers, traders, banks and buyers—are being formed. All members will work on a common platform, on which records of every successful transaction and contract are tokenised in a chain of blocks, on view for all to see. This transparency

enables bankers to finance an operation with greater confidence, buyers to check the due diligence on a large supply and oil commodity traders to get more transparent.

This is the model that the electricity utility sector is looking at—in particular, the renewable energy segment.

Limited in scope?

The renewable energy market in the US is currently estimated to be US\$ 7 billion. This suggests that blockchain-based trading has a vast scope here. Yet, scaled-out models are few, at the moment. Blockchain + IoT in the utility space appears to still be a work-in-progress.

In March 2016, Massachusetts Institute of Technology (MIT) Enterprise Forum of Cambridge, Massachusetts, played host to an interesting presentation. Ewald Hesse, chief executive officer, Grid Singularity—Austria-based startup in the blockchain space—was on stage. He presented a unique case study of blockchain usage.

In Soweto, a poor outer suburb of Johannesburg, South Africa, most homes are off-grid. Lorien Gamaroff, chief executive officer, Bankymoon—South African blockchain firm—decided to step in and somewhat adopt a local school. Bankymoon set up a smart meter at the school that is programmed to accept digital currency. This meter is also linked to a platform where donors can donate Bitcoin. As long as the cryptocurrency has not all been spent, the local utility supplies the school with electricity. When the currency runs out, the school goes off-grid.

Across the world, at Boston, Hesse shared his video call with Gamaroff with the audience at MIT Enterprise Forum. On the large screen on an MIT stage, the distant Soweto school was seen to be in darkness. And, when someone from Boston donated one Bitcoin on Bankymoon's platform, the school lit up to the sound of applause from the children in Soweto.

This pilot shows one way that blockchain can be used by the donor community. This would involve no middlemen, nor high overheads of organisations like Oxfam, United Nations and Red Cross, where administration costs are known to eat up

about 30 per cent of donations, and often even more.

The way forward

As Singh mentioned in his *IEW* talk, the way forward lies in forming large consortiums, with many partners from different levels of a sector's ecosystem, working on a common platform. Grid Singularity has tied up with the well-respected US-based NGO, Rocky Mountain Institute, to form Energy Web Foundation. This has many partners and users, like Shell, ShareCharge, Wipro and a host of other big utilities and blockchain solution providers. If this platform gets more accepted and signs on partners from across the globe, the interesting PoCs and pilots being demonstrated in isolated pockets around the world could probably scale out efficiently, truly demonstrating how blockchain could save millions of dollars in cutting out middlemen.

In the Indian context, energy distribution is entirely in the hands of the government and, hence, locked up in bureaucratic red tape. Apart from that, distribution infrastructure is dated. In response to an audience-member's question about India's readiness for blockchain application, Singh said, "In India, we are still dealing with electricity theft." Sadly, the country is struggling with that, as well as electricity loss due to inefficient systems.

Further, getting all the country's states onto a common platform itself would pose quite a challenge. And then, upgrading its vast distribution network to the level of smartness required to enable blockchain usage lies many years in the future.

If at all opportunities exist for blockchain application in the Indian energy space, it could be in renewable energy alone, and possibly in vast gated technology parks or housing estates, where smartness has already been built into the infrastructure. Some examples are smart access control, smart surveillance and IoT usage to ensure efficient energy usage. With such infrastructure as a base, it would be possible to replicate the Brooklyn Grid experiment in India. **EFY**

This article was triggered by a talk given by Prashant K. Singh, senior consultant, Blockchain Centre of Excellence, Wipro at India Electronics Week 2018